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Data-Driven Analysis of European Flight Delays

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Abstract

Every year, flight delays impose a multi-billion-euro burden on the European aviation industry, but the true cost is measured in more than just currency. For passengers, a delay often means a missed connection or a lost day; for airlines, it translates to wasted fuel and a 'domino effect' of scheduling disruptions that can paralyze entire airport networks. While the complexity of these variables makes forecasting a significant challenge, the ability to predict delays is essential for a more resilient sky. This study introduces a comprehensive analytical framework designed to predict flight delays across five of Europe's most critical aviation hubs: the United Kingdom, Germany, France, the Netherlands, and Spain.

We use only public data sources. The plan is to combine airport information from the OurAirports project with airport-level punctuality records published by EUROCONTROL. After that, we will create at least fifteen useful features from the raw data and build one clean dataset containing more than one thousand rows. Using this dataset, we will train and compare several supervised learning models, including logistic regression, decision tree, and random forest. We will also report the correct evaluation metrics and explain what the results actually show.

The goal of this project is not to outperform advanced research papers. Many of the studies we review later use deep neural networks, gradient boosting models, and much larger datasets than ours. Competing with that is not our purpose. Our goal is to build a pipeline that is clear, reproducible, honest, and easy for another student or reader to follow, while staying within the limits of an undergraduate course. Choosing five countries that all report through the same regulator, EUROCONTROL, helps with that because it avoids many common problems where the same column means different things in different files.

1. Introduction

A delayed flight looks simple from a passenger's seat. You sit at the gate, the screen says "delayed" and eventually you board. From the airline's side though, that single delay is the visible tip of a much messier and complicated process. Crew shifts, aircraft rotations, ground handling, weather at airports thousands of kilometers away, air-traffic-flow restrictions, runway capacity all of these feed into whether a particular flight leaves on time. International regulators have been trying to measure punctuality for decades, and the metric that has been used the most is **OTP15**: a flight counts as on time if it arrives or departs within fifteen minutes of its schedule. EUROCONTROL and most national civil aviation authorities still report on this fifteen-minute basis.

Why bother predicting delays at all? Two reasons basically. The practical one is that anyone who can predict tomorrow's punctuality with even modest accuracy can plan better gate assignments, fuel loads, crew rotations, and passenger communication. The scientific one is that flight delay is a textbook hard prediction problem. The data is messy and comes from many sources. The unit of analysis is genuinely not clear (do you predict per flight? per airport? per route per month?). The target is imbalanced and not certain because most flights at well-run airports actually are on time. Even deciding what "the target" is becomes a research decision rather than a given.

Our project takes a cross-country angle. Most of the literature we read sticks to one airport or one country, often the United States, because the U.S. Bureau of Transportation Statistics publishes the richest free flight-level dataset in the world. That convenience is also a limitation: a model trained on Atlanta and Chicago does not necessarily say much about Madrid in August or Schiphol in fog. That's why we deliberately focus on five European markets which are the UK, Germany, France, the Netherlands, and Spain, chosen to cover the busiest hubs on the

continent (Heathrow, Frankfurt, Munich, Paris–Charles-de-Gaulle, Schiphol, Madrid-Barajas, Barcelona-El Prat) while still sharing one regulator and one delay definition. In other words, we get geographic and operational variety without paying the price of incompatible data formats.

On top of EUROCONTROL's airport-day records we layer airport metadata from OurAirports: type (large/medium/small), latitude, longitude, elevation, region, municipality, and a flag for whether the airport runs scheduled service. These fields let us derive richer features later (for example, hub status or distance from a coast), and they make sure we hit the brief's requirement of at least fifteen meaningful features once temporal and engineered features are added.

The five countries also matter for what they capture between them. Northern Europe brings winter-weather disruption (de-icing, low visibility). The Netherlands brings the runway-capacity problem at Schiphol. France brings ATFM strikes as a known outlier source. Spain brings extreme summer-tourism seasonality and a mix of mainland, Mediterranean, and Atlantic-island airports that no other single country in our set provides. So even though the dataset is "small" by industry standards, it is structurally diverse and is different from each other which is enough to be worth modelling.

By the end of this report (and the rest of the project) the question we want to answer is simple: given what we know about an airport-day before it happens, the country, the airport, the season, the weekday, the planned traffic volume, how well can we predict whether punctuality will hold up? Phase 2 is where we set up the background, the domain, and make this report; Phases 3 and 4 deliver the actual answer.

2. Background

2.1 What 'delay' actually means

It sounds obvious, but it isn't. The word "delay" in everyday speech covers at least three different operational definitions and isn't a single meaning, and a paper that doesn't make the distinction explicit is hard to trust, that's why we will cover the word.

Departure delay: is the difference between when an aircraft actually pushes back from the gate and when it was scheduled to push back.

Arrival delay is the same idea at the destination: actual on-block time minus scheduled on-block time. Both are signed numbers in minutes; a flight can be early (negative delay) too, although in practice early arrivals are rare and small.

OTP15 punctuality: the third definition, is a binary flag built on top of those minutes. A flight is "on time" if its departure or arrival delay is fifteen minutes or less. This is the metric you see in monthly EUROCONTROL bulletins and in airport press releases. It is also the most useful one for a classification model because it converts a noisy continuous quantity into an easy and clean yes or no label.

Why does this matter for us? Because the same EUROCONTROL data we are using can drive two very different machine-learning tasks. We could regress on the average schedule delay (in minutes per flight per airport-day) and treat the problem as a continuous one. Or we could categorize it, predict whether the airport-day fails an OTP15 target, and treat it as binary classification. We are deliberately keeping both options open in Phase 2, and we will commit during Phase 3 once we have actually looked at the cleaned data.

2.2 Causes of flight delays

Years of regulator reports and academic papers all converge on roughly the same three types of delay causes.

operational: things that aren't fully in the airlines control but is part of the airline operations. Examples being crew showing up late, maintenance issues, slow baggage loading, and (the big one) delays propagating from a previous leg of the same aircraft's day. If your 09:00 flight from Frankfurt to Madrid leaves twenty minutes late, the same plane's 12:30 from Madrid to Lisbon is probably going to leave late too. Delay propagation is famously the largest single contributor at hub airports.

Environmental: which is basically weather. Wind, low visibility, thunderstorms, snow, fog. These hit different airports in different ways. Northern hubs have winters; Mediterranean hubs have summer thunderstorms and capacity overload; coastal hubs have wind direction issues for runway choice.

The third is **network and capacity**. Air-traffic flow management (ATFM) regulations, airspace congestion, runway closures, and peak-hour bottlenecks. EUROCONTROL's CODA reports often attribute large slices of European delay minutes to ATC capacity at specific en-route control centers, and to occasional regulator strikes (France and Spain in particular) that produce dramatic but localized spikes.

If you build a model that ignores even one of those three types, you are leaving real important details on the floor. We try to capture all three indirectly through features that are observable *before* an airport-day operates: planned traffic volume, country, season, day of week, airport size class, and (where we can get it) airport geographic features like elevation and coastal proximity.

2.3 Why this is a hard machine-learning problem

Three things make naive or basic modelling on this data unreliable, and they all can do a harmful effect if you don't notice them.

class imbalance: Most airport-days at well-managed European airports do meet their OTP15 target. Days that fail are a minority about 20–25% of the records, depending on the airport and year. A classifier that always predicts "on target" would clock around 80% accuracy and be completely useless in practice. So, we need to watch F1, ROC-AUC, and Matthews Correlation Coefficient (MCC) instead of accuracy, and we may need to resample the training data to give the minority class a fair chance.

non-stationarity: The aviation system in 2019 is not the aviation system in 2024. COVID affected European traffic, the rebound was uneven, schedules and fleets reshuffled, and ATM procedures changed. A model trained naively across that boundary can pick up patterns that no longer hold. We will deal with this by being careful about which years we include and by using a temporal train/test split rather than a random one.

delay propagation: we already mentioned it above. A flight that is late at 09:00 in Madrid creates a late aircraft at 11:00 in Paris which creates a late connection at 14:00 in Amsterdam. This temporal-and-network coupling is exactly what graph neural networks and BiLSTMs in the recent literature try to model directly. We are not going there. We are doing a project working at the airport-day level, and we will handle propagation indirectly through engineered features (day of week, season, hub flag, traffic-volume class) rather than by building an explicit temporal model.

3. Domain Explanation

3.1 The aviation data ecosystem

Public aviation data works in two layers, and most useful projects join them together and they are important to understand to make a good project.

The **metadata layer** describes airports, airlines, and aircraft with their properties that change slowly or not at all. Our project use OurAirports which is hosted as open data and get updated daily also being a free source system. Each row is one airport with its IATA and ICAO codes, type (Examples : large_airport, medium_airport, small_airport, heliport, seaplane_base, balloonport, closed), ISO country and region codes, latitude, longitude, elevation in feet, municipality, and a flag for whether the airport runs scheduled service. Academic and industry projects use OurAirports constantly, and there is no good reason to replace it.

The **performance layer** describes what actually happened on a given day or flight: scheduled and actual times, cancellation and diversion flags, and (in some sources) the cause of delay. Our performance source is **EUROCONTROL Aviation Intelligence Portal - Airport Punctuality**. EUROCONTROL publishes daily and monthly punctuality (OTP15) and average schedule delay metrics for European airports. It has two units of analysis we are going to use which are the airport-day or airport-month, not the individual flight, because for our scope that level of detail in a set of data is more than enough. With multiple years times multiple airports times multiple countries, we comfortably exceed the 1,000-row requirement.

We chose to use only one performance source which has two benefits. The first is that all five countries appear in the same file with the same column names and the same definitions, so we don't have to write country-specific logic or argue about whether "delay" means the same thing in row 1 and row 1000 because here delay is the same for all countries covered in the source. The second is that the unit of analysis is consistent across countries, which means the cross-country comparison we want to do later is actually meaningful we are not silently comparing different things.

3.2 Punctuality definitions used by regulators

Three definitions show up over and over in the sources we are using and in our report, and they are important of this report so it needs to be explained.

OTP15 (On-Time Performance, fifteen-minute threshold): A flight is on time if it departs or arrives within 15 minutes of schedule. EUROCONTROL reports network-wide OTP15 monthly. As an order of magnitude figure, EUROCONTROL recently reported network arrival punctuality of roughly 76% across more than eleven million flights in 2025.

Average schedule delay (minutes per flight): Just the mean of (actual scheduled) times across the relevant population and can be computed (per airport per day or per month). This number is the natural target for a regression model because it preserves the magnitude of the delay and not just its yes or no nature.

IATA delay codes: A two-digit numeric system where each minute of delay is attributed to one of dozens of named categories (Examples: passenger handling, fuelling, baggage, ATFM weather, ATFM capacity, cabin damage, late aircraft, and so on). EUROCONTROL's CODA reports group these codes into broader operational families. We probably will not use the codes directly because EUROCONTROL's airport-day file does not break them down per record, but they are essential context for reading anything on this topic.

3.3 Geographic and policy context for our five countries

We did not pick the five countries randomly. Each of them is a major European market, and is well-covered in EUROCONTROL data, and together they can have enough variation in climate, traffic mix, and operational profile to make the modelling interesting.

United Kingdom: Heathrow, Gatwick, Manchester, Stansted. Heathrow is one of the most capacity-constrained large airports in the world its punctuality is dominated by what happens when a single runway closes. The UK also gets the full North Atlantic winter weather story, with low visibility and storm disruption between roughly November and March.

Germany: Frankfurt and Munich are both top-ten European airports by traffic. Frankfurt is a Lufthansa mega-hub, its delays propagate aggressively across the European network because so many connections route through it. German airports are also unusual in how tightly they integrate with the ICE high-speed rail network, which interacts with airline scheduling in ways different than most countries.

France: Paris–Charles-de-Gaulle, Paris-Orly, Nice, Lyon. French airports are particularly exposed to ATFM disruption during national air traffic controller industrial action a recurring source of large but sharply-localised delay spikes. Any model we build has to be ready to treat those days as outliers and unusual data rather than typical and normal behaviour, otherwise it will overfit to dramatic events and underpredict normal ones.

Netherlands: Almost everything happens at Amsterdam Schiphol, which is one of Europe's densest hubs for its physical footprint. Schiphol's punctuality is famously sensitive to runway-capacity management and to North-Sea weather; even a moderate crosswind forces runway reallocation and pushes delays up.

Spain: This is the most geographically diverse country in our set. Mainland hubs (Madrid-Barajas, Barcelona-El Prat) sit alongside Mediterranean coastal airports (Palma de Mallorca, Alicante, Málaga, Valencia) and Atlantic-island airports in the Canaries (Tenerife, Gran Canaria). Spain has the strongest summer-tourism seasonality of the five countries by a clear margin, which makes derived temporal features like month, season, and holiday flag genuinely informative on Spanish airport-days. The latitude and elevation spread is also wider than any other country in our list, which helps the metadata-driven part of the feature set.

The fact that all five countries report through EUROCONTROL using the same OTP15 and average-delay definitions is perfect with the sources we are using. Without it, "comparing" the UK and Spain would be comparing whatever each country's national authority chose to publish, and the comparison would tell us more about reporting behaviour than about real differences in performance.

4. Algorithm Background

This section walks through the families of supervised learning algorithms we plan to compare in Phase 4. We are deliberately staying inside what was taught in the data science course and inside the libraries declared in ``requirements.txt``.

4.1 Logistic regression

The classical baseline for binary classification. It models the log-odds of the positive class as a linear combination of input features, fits the coefficients by maximum likelihood, and turns those log-odds into a probability through the sigmoid function. Logistic regression is interpretable in a way that more complicated models are not: each coefficient maps directly onto how much one feature changes the log-odds of being delayed. When the relationship between features and target is approximately linear (or can be made so by sensible feature engineering), it is also surprisingly competitive. In the flight-delay literature, it is the standard baseline that more complex models have to justify themselves against.

4.2 Decision trees

A single decision tree partitions the feature space by recursively choosing the split that most reduces impurity. For classification we measure impurity with Gini or entropy; for regression we use variance reduction. Trees handle non-linearities and feature interactions naturally, work fine with mixed numeric and categorical inputs after encoding, and are easy to visualise. Their main weakness is instability, a small change in the training data can produce a structurally different tree, and an unconstrained tree will overfit aggressively. We expect a single tree to underperform the random forest, but it is worth fitting one as a sanity check and to extract feature importance for interpretation.

4.3 Random Forest

This is our default expected best model for this dataset. A random forest is a bagged ensemble of decision trees. Each tree is trained on a bootstrap sample of the training set, and at each split it considers only a random subset of the available features. The final prediction averages (regression) or votes (classification) across all the trees. This randomisation kills the instability problem of single trees and reduces variance dramatically. Random forests are robust to noise, almost free of hyper-parameter sensitivity beyond ``n_estimators`` and ``max_depth``, and they

hand you a feature-importance ranking for free. The literature we review repeatedly reports random forest as either the best model or a strong runner-up on flight-delay tasks, especially when the feature set is heterogeneous (categorical airport codes mixed with continuous geographic and temporal features). That fits our setup almost perfectly.

4.4 Gradient boosting (note rather than commitment)

We mention gradient boosting (XGBoost, LightGBM, CatBoost) because it is impossible to read the recent flight-delay literature without it, it is repeatedly the highest-scoring model in published comparisons, including the 2024 paper by Hatipoğlu and Tosun reviewed below. Conceptually, gradient boosting builds an ensemble sequentially, with each new weak learner fitted to the residuals of the current ensemble. The gain in accuracy over random forest can be real, but the libraries that implement it well (XGBoost, LightGBM, CatBoost) are not part of the basic scikit-learn distribution and may fall outside Section 7 of the brief. Our default position is to leave them out unless explicitly approved.

4.5 Evaluation metrics

For a binary "on-target / not-on-target" classification target, we will report:

- Accuracy, but only as a sanity check and we will call out explicitly that on an imbalanced dataset accuracy alone is misleading.
- Precision, recall, and F1-score for the positive ("missed-target") class, because what we actually care about is catching the bad days.
- ROC-AUC, which summarises the trade-off between true-positive rate and false-positive rate at every classification threshold and is the cleanest single number on imbalanced data.
- A confusion matrix, because looking at where the model is actually getting things wrong almost always teaches us something the aggregate metrics miss.

For a regression target (average delay in minutes per airport-day), we will report Mean Absolute Error and Root Mean Squared Error. MAE is more interpretable for a non-technical reader; RMSE penalises large errors more aggressively and matches what most of the literature reports.

We split data temporally rather than randomly. Random splits on time-correlated data give optimistically biased scores because the model can see the "future" in training. A temporal split (train on 2022, test on 2023, say) is closer to how the model would actually be deployed.

5. Review of Related Work

We picked five recent papers on flight delay prediction, all from the last five years. The selection deliberately focus on different geographies, data sets details, and method families so that our own design choices can be argued against a representative sample of the field not just a convenient subset. One of the five papers (Silvestre et al., 2024) is set at Madrid-Barajas, which is in our country list, so it is a particularly direct point of comparison.

5.1 Dai (2024)- Hybrid clustering plus per-cluster learning at JFK

Dai (2024), in *Scientific Reports*, proposes a two-stage pipeline. In stage one, flights are grouped into clusters by similarity. In stage two, a separate predictive model is trained for each cluster, on the theory that one model can't fit the heterogeneous mix of flight types arriving at a major hub equally well. The dataset is 22,796 arrival records at New York-JFK from January to August 2023, scraped from FlightRadar24. The class distribution is heavily imbalanced: most flights are on time, a minority are mildly delayed, and a tiny fraction are significantly delayed. Dai shows that the clustered approach beats single stand-alone gradient-boosting baselines and suggests the gain comes from letting each cluster have its own decision boundary.

For us the take-away is not the clustering trick (we are not doing that), it is the imbalance lesson. Class imbalance is the single most consistent pitfall in flight-delay classification. If we report only accuracy and ignore minority-class recall, we will be reporting a model that looks good and is operationally useless.

5.2 AlBassam and AlShahrani (2025)- Resampling on a Kaggle dataset

AlBassam and AlShahrani (2025), in *PLOS ONE*, work on the publicly available "Flight Status Prediction" dataset on Kaggle about four million records from 2018 to 2022. They benchmark several classical classifiers (Random Forest, Decision Tree, Naive Bayes, and a few others) and run a careful comparison of resampling techniques: Random Oversampling, Random Undersampling, and SMOTE. The best combinations reach 0.90 accuracy and 0.90 F1 on the binary delayed/on-time target.

What we like about this paper is two things. First, it makes the methodologically important point that a simple probabilistic classifier like Naive Bayes can actually be competitive once the data is balanced the assumption that more complexity always helps is just wrong. Second, the paper is the closest in spirit to our own intended approach: classical supervised models, careful evaluation across multiple metrics, and an explicit treatment of imbalance. We are not going to re-implement their pipeline blindly, but their structure is a reasonable template.

5.3 Bisandu and Moulitsas (2023)- Deep BiLSTM on U.S. on-time data

Bisandu and Moulitsas (2023), in the **Journal of Aviation/Aerospace Education & Research**, train Bidirectional Long Short-Term Memory (BiLSTM) and standard LSTM models on U.S. Bureau of Transportation Statistics on-time-performance data. After feature selection driven by three explicit criteria, the BiLSTM reaches a Matthews Correlation Coefficient of 0.99 on the held-out test set, comfortably beating the single-direction LSTM.

A 0.99 MCC is an eye-catching number, and we read it with a healthy dose of scepticism. Extreme correlation values in flight-delay datasets very often signal feature leakage , example: a feature that encodes the actual departure time accidentally being included as an input when predicting arrival delay. We cite this paper anyway because it is a useful reference point for the **upper end** of what deep models claim on flight-level data, and our own much simpler models (working on coarser airport-day data) should be benchmarked against that ceiling honestly, not pretended-against.

5.4 Hatipoğlu and Tosun (2024)- XGBoost with weather and SHAP at a Turkish airport

Hatipoğlu and Tosun (2024), in **Applied Sciences**, predict whether arrival flights at a Turkish airport will be more than fifteen minutes late, using three years of operational data joined with meteorological observations. They run a long bench: Logistic Regression, Naive Bayes, Neural Networks, Random Forest, XGBoost, CatBoost, and LightGBM. They use SMOTE to deal with imbalance and SHAP values to interpret the resulting models. XGBoost wins at roughly 80% accuracy.

The most interesting result in this paper is actually a negative one. Adding weather features did *not* materially improve performance over the operational features alone. This is a useful caution against the lazy assumption that "of course weather predicts delay". It does, but only as much as the operational features have not already absorbed. If you already know the time of day, the airport, the carrier, and the season, weather may add less than you expect. We will keep this in mind in Phase 3 when we decide whether to spend effort fetching weather summaries. The Turkish setting (a busy Mediterranean airport with a mixed tourist-and-business traffic profile) is also a useful informal analogue for Spanish coastal airports in our own dataset.

5.5 Silvestre et al. (2024)- LSTM for arrival time prediction at Madrid-Barajas

Silvestre, Martínez-Prieto, Bregon, and Álvarez-Esteban (2024), in **The Journal of Supercomputing**, use LSTM networks that combine a flight's four-dimensional trajectory (latitude, longitude, altitude, time) with destination-airport weather data to predict in-flight estimated time of arrival. The model is evaluated on flights arriving at **Adolfo Suárez Madrid-Barajas Airport (Spain)** during the first three quarters of 2022, drawing on data from forty origin airports. They report a mean absolute error of 2.65 minutes and a root-mean-squared error of 4.30 minutes across the entire flight, which beats the ensemble and boosting baselines they compare against.

This paper is the most directly relevant of the five for us. It is set at one of the airports we will be modelling. It confirms that, with enough features, ETA-style targets at a Spanish hub are predictable to within a few minutes, although it does so at flight level with a deep model, not at airport-day level with classical methods. It also makes the case for joining operational features with weather features, which is exactly the design pattern we are considering for Phase 3 (EUROCONTROL operational metrics joined with OurAirports metadata, and possibly with simple weather-summary fields if we can find a clean public source).

5.6 What we take from these five papers

Within these five papers we find three themes and patterns being consistent on them.

Tree-based ensembles are the workhorse of this field: Across U.S., Saudi-Kaggle, Turkish, and Spanish datasets, random forest and gradient boosting are repeatedly either the winning model or close to it. Even when deep models technically score higher, the gap is often small enough that the simpler models win on being understandable and training cost.

Imbalance handling matters more than architecture: A logistic regression with SMOTE will often beat a fancier model that ignores class imbalance. The papers that explicitly use resampling techniques systematically report better minority-class recall than those that don't. We will follow this lesson directly.

Single-airport scope is the norm, and it is also a known weakness: Dai uses JFK only. Silvestre et al. use Madrid-Barajas only. Hatipoğlu and Tosun use one Turkish airport. The papers that work on multiple airports almost always still stay inside one country. Our project's five-country, multi-airport European scope is a genuine contribution. It forces our cleaning and feature-engineering pipeline to be portable across airports with very different traffic mixes, winter-weather hubs versus summer-tourism hubs, mega-hubs versus regional airports without country-specific hard-coding.

We are not trying to create a new algorithm. We are trying to show that a transparent, classical machine-learning pipeline, built on one consistent regulator's data, can produce honest and useful predictions across five European countries, and we want the report to read like one coherent argument rather than a stack of disconnected experiments.

References

AlBassam, S. A. A., & AlShahrani, D. N. (2025). Flight delay prediction: Evaluating machine learning algorithms for enhanced accuracy. **PLOS ONE, 20*(12)*, e0335141.

<https://doi.org/10.1371/journal.pone.0335141>

Bisandu, D. B., & Moulitsas, I. (2023). A deep BiLSTM machine learning method for flight delay prediction classification. **Journal of Aviation/Aerospace Education & Research, 32*(2)*, Article 4.

<https://doi.org/10.58940/2329-258X.1992>

Dai, M. (2024). A hybrid machine learning-based model for predicting flight delay through aviation big data. **Scientific Reports, 14**, 4603.

<https://doi.org/10.1038/s41598-024-55217-z>

EUROCONTROL. (2026). **Data Snapshot #57 on 2025 European aviation in numbers.**

<https://www.eurocontrol.int/publication/eurocontrol-data-snapshot-57-2025-european-aviation-numbers>

Hatipoğlu, I., & Tosun, Ö. (2024). Predictive modeling of flight delays at an airport using machine learning methods. **Applied Sciences, 14*(13)*, 5472.

<https://doi.org/10.3390/app14135472>

OurAirports. (2026). **Open data downloads.**

<https://ourairports.com/data/>

(mirror: <https://davidmegginson.github.io/ourairports-data/>)

Silvestre, J., Martínez-Prieto, M. A., Bregon, A., & Álvarez-Esteban, P. C. (2024). A deep learning-based approach for predicting in-flight estimated time of arrival. **The Journal of Supercomputing, 80*(12)*, 17212–17246.

<https://doi.org/10.1007/s11227-024-06060-6>